Term 3 Robotics Challenge

Interim Report

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1. Mechanical Design [10 pts]

1.1. General Project Outline [2 pts]

- Project development timeline for the 6 weeks of the Challenge, with milestones.
- A full bill of **mechanical components and consumables** (at this stage of the project).

1.2. Theory and Background [3 pts]

- What mechanism(s)/approach/robot design did you choose for the main sub-systems of your robot? What was the motivation behind these design ideas?
- Relevant **foundation** behind your design:
  - A schematic diagram of the robot with clear labels of all parameters and variables/symbols.
  - Description of how your robot operates.

1.3. Mechanical Design and Manufacturing [5 pts]

- **CAD models** of the most important parts and blocks of your robot.
- **Exploded view** of the robot's assembly.
- **Photos** of the various stages of your robot manufacturing and assembly process (*laser cutting, 3D printing, workshop, testing etc.*)
- A photo of the **assembled prototype** (mechanical part).

2. Electrical Design [10 pts]

2.1. Conceptual Design [4 pts]

- A **schematic diagram** of your system with clear labels of all parameters and variables/symbols.
- Description of how the main electronic systems/blocks of your **robot operate** (*the purpose of main components/blocks and how they integrate*).
- Description of data **signals and waveforms** (*from MCU to motors, from sensors to MCU, etc.*).

2.2. Practical Implementation [6 pts]

- A full **bill** of electrical and electronic components.
- Provide **photos** of assembly/manufacturing steps of the real circuit/system (*e.g. soldering, connecting wires*).
- A photo of a **complete, assembled electrical sub-system** of your robot.
- Photos/screenshots/plots or other results that would demonstrate your **electronics in action** (*e.g. screenshots of Arduino IDE Serial Monitor that demonstrate changes in signals from robots' reflectance and distance sensors*).
- Feel free to add any other relevant details to help with your story-telling.